

Bellabeat Capstone Project – A Data-Driven Wellness Story

MUTHUSRIHEMADHARSHINI S A

2025-06-02

Introduction

Business Task:

Analyze trends in smart device usage using Fitbit data.

Identify behavioral patterns in sleep, activity, and calorie burn,

then apply these insights to support marketing strategy for the Bellabeat app and products.

```
install.packages('tidyverse')
```

Install and load the tidyverse

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.4'  
## (as 'lib' is unspecified)
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
## v dplyr      1.1.4      v readr      2.1.5  
## v forcats    1.0.0      v stringr    1.5.1  
## v ggplot2    3.5.2      v tibble     3.2.1  
## v lubridate  1.9.4      v tidyr      1.3.1  
## v purrr      1.0.4
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()      masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
daily_activity <- read_csv("dailyActivity_merged.csv")
```

Loading datasets and creating dataframes

```
## Rows: 940 Columns: 15
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr  (1): ActivityDate
```

```
## dbl (14): Id, TotalSteps, TotalDistance, TrackerDistance, LoggedActivitiesDi...
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
sleep_day <- read.csv("sleepDay_merged.csv")
weight_info <- read.csv("weightLogInfo_merged.csv")
hourly_steps <- read.csv("hourlySteps_merged.csv")
heart_rate <- read.csv("heartrate_seconds_merged.csv")
mets_data <- read.csv("minuteMETsNarrow_merged.csv")
```

```
daily_activity <- distinct(daily_activity)
sleep_day <- distinct(sleep_day)
weight_info <- distinct(weight_info)
hourly_steps <- distinct(hourly_steps)
heart_rate <- distinct(heart_rate)
mets_data <- distinct(mets_data)
```

Taking only distinct values by removing redundant data

```
daily_activity$ActivityDate <- mdy(daily_activity$ActivityDate)
sleep_day$SleepDay <- mdy_hms(sleep_day$SleepDay)
sleep_day$SleepDay <- as_date(sleep_day$SleepDay)
weight_info$Date <- mdy_hms(weight_info$Date)
weight_info$Date <- as_date(weight_info$Date)
hourly_steps$ActivityHour <- ymd_hms(hourly_steps$ActivityHour)
```

Formatting Dates

```
## Warning: All formats failed to parse. No formats found.
```

```
heart_rate$Time <- ymd_hms(heart_rate$Time)
```

```
## Warning: All formats failed to parse. No formats found.
```

```
mets_data$ActivityMinute <- ymd_hms(mets_data$ActivityMinute)
```

```
## Warning: All formats failed to parse. No formats found.
```

```
head(daily_activity)
```

Taking a look at our data

```
## # A tibble: 6 x 15
##       Id ActivityDate TotalSteps TotalDistance TrackerDistance
##       <dbl> <date>         <dbl>         <dbl>         <dbl>
## 1 1503960366 2016-04-12         13162           8.5           8.5
## 2 1503960366 2016-04-13          10735           6.97          6.97
## 3 1503960366 2016-04-14          10460           6.74          6.74
## 4 1503960366 2016-04-15           9762           6.28          6.28
## 5 1503960366 2016-04-16          12669           8.16          8.16
## 6 1503960366 2016-04-17           9705           6.48          6.48
## # i 10 more variables: LoggedActivitiesDistance <dbl>,
## #   VeryActiveDistance <dbl>, ModeratelyActiveDistance <dbl>,
## #   LightActiveDistance <dbl>, SedentaryActiveDistance <dbl>,
## #   VeryActiveMinutes <dbl>, FairlyActiveMinutes <dbl>,
## #   LightlyActiveMinutes <dbl>, SedentaryMinutes <dbl>, Calories <dbl>
```

```
head(sleep_day)
```

```
##           Id  SleepDay TotalSleepRecords TotalMinutesAsleep TotalTimeInBed
## 1 1503960366 2016-04-12                1                327             346
## 2 1503960366 2016-04-13                2                384             407
## 3 1503960366 2016-04-15                1                412             442
## 4 1503960366 2016-04-16                2                340             367
## 5 1503960366 2016-04-17                1                700             712
## 6 1503960366 2016-04-19                1                304             320
```

```
head(weight_info)
```

```
##           Id      Date WeightKg WeightPounds Fat   BMI IsManualReport
## 1 1503960366 2016-05-02    52.6    115.9631  22 22.65             True
## 2 1503960366 2016-05-03    52.6    115.9631  NA 22.65             True
## 3 1927972279 2016-04-13   133.5    294.3171  NA 47.54            False
## 4 2873212765 2016-04-21    56.7    125.0021  NA 21.45             True
## 5 2873212765 2016-05-12    57.3    126.3249  NA 21.69             True
## 6 4319703577 2016-04-17    72.4    159.6147  25 27.45             True
##           LogId
## 1 1.462234e+12
## 2 1.462320e+12
## 3 1.460510e+12
## 4 1.461283e+12
## 5 1.463098e+12
## 6 1.460938e+12
```

```
head(hourly_steps)
```

```
##           Id ActivityHour StepTotal
## 1 1503960366      <NA>         373
## 2 1503960366      <NA>         160
## 3 1503960366      <NA>         151
## 4 1503960366      <NA>           0
## 5 1503960366      <NA>           0
## 6 1503960366      <NA>           0
```

```
head(heart_rate)
```

```
##           Id Time Value
## 1 2022484408 <NA>    97
## 2 2022484408 <NA>   102
## 3 2022484408 <NA>   105
## 4 2022484408 <NA>   103
## 5 2022484408 <NA>   101
## 6 2022484408 <NA>    95
```

```
head(mets_data)
```

```
##           Id ActivityMinute METs
## 1 1503960366      <NA>    10
## 2 1503960366      <NA>    10
## 3 1503960366      <NA>    10
## 4 1503960366      <NA>    10
## 5 1503960366      <NA>    10
## 6 1503960366      <NA>    12
```

```
colnames(daily_activity)
```

```
## [1] "Id" "ActivityDate"
## [3] "TotalSteps" "TotalDistance"
## [5] "TrackerDistance" "LoggedActivitiesDistance"
## [7] "VeryActiveDistance" "ModeratelyActiveDistance"
## [9] "LightActiveDistance" "SedentaryActiveDistance"
## [11] "VeryActiveMinutes" "FairlyActiveMinutes"
## [13] "LightlyActiveMinutes" "SedentaryMinutes"
## [15] "Calories"
```

```
colnames(sleep_day)
```

```
## [1] "Id" "SleepDay" "TotalSleepRecords"
## [4] "TotalMinutesAsleep" "TotalTimeInBed"
```

```
colnames(weight_info)
```

```
## [1] "Id" "Date" "WeightKg" "WeightPounds"
## [5] "Fat" "BMI" "IsManualReport" "LogId"
```

```
colnames(hourly_steps)
```

```
## [1] "Id" "ActivityHour" "StepTotal"
```

```
colnames(heart_rate)
```

```
## [1] "Id" "Time" "Value"
```

```
colnames(mets_data)
```

```
## [1] "Id" "ActivityMinute" "METs"
```

Understanding some summary statistics

How many unique participants are there in each dataframe?

```
n_distinct(daily_activity$Id)
```

It looks like there may be more participants in the daily activity dataset than the sleep dataset.

```
## [1] 33
```

```
n_distinct(sleep_day$Id)
```

```
## [1] 24
```

```
nrow(daily_activity)
```

How many observations are there in each dataframe?

```
## [1] 940
```

```
nrow(sleep_day)
```

```
## [1] 410
```

Quick summary statistics about each data frame?

```
daily_activity %>%
  select(TotalSteps,
         TotalDistance,
         SedentaryMinutes) %>%
  summary()
```

For the daily activity dataframe:

```
##      TotalSteps    TotalDistance    SedentaryMinutes
##  Min.      :    0    Min.      : 0.000    Min.      :    0.0
## 1st Qu.: 3790    1st Qu.: 2.620    1st Qu.: 729.8
## Median : 7406    Median : 5.245    Median :1057.5
## Mean   : 7638    Mean   : 5.490    Mean    : 991.2
## 3rd Qu.:10727    3rd Qu.: 7.713    3rd Qu.:1229.5
## Max.   :36019    Max.   :28.030    Max.    :1440.0
```

```
sleep_day %>%
  select(TotalSleepRecords,
         TotalMinutesAsleep,
         TotalTimeInBed) %>%
  summary()
```

For the sleep dataframe:

```
## TotalSleepRecords TotalMinutesAsleep TotalTimeInBed
## Min.      :1.00      Min.      : 58.0      Min.      : 61.0
## 1st Qu.:1.00      1st Qu.:361.0      1st Qu.:403.8
## Median :1.00      Median :432.5      Median :463.0
## Mean   :1.12      Mean   :419.2      Mean   :458.5
## 3rd Qu.:1.00      3rd Qu.:490.0      3rd Qu.:526.0
## Max.   :3.00      Max.   :796.0      Max.   :961.0
```

```
weight_info %>%
  select(WeightKg,
         WeightPounds,
         BMI) %>%
  summary()
```

For the weightLogInfo

```
##      WeightKg      WeightPounds      BMI
##  Min.      : 52.60    Min.      :116.0    Min.      :21.45
## 1st Qu.: 61.40    1st Qu.:135.4    1st Qu.:23.96
## Median : 62.50    Median :137.8    Median :24.39
## Mean   : 72.04    Mean   :158.8    Mean   :25.19
## 3rd Qu.: 85.05    3rd Qu.:187.5    3rd Qu.:25.56
## Max.   :133.50    Max.   :294.3    Max.   :47.54
```

```
mets_data %>%
  select(Id,
         METs)%>%
  summary()
```

For mets_data

```
##           Id           METs
## Min.      :1.504e+09   Min.   : 0.00
## 1st Qu.:2.320e+09   1st Qu.: 10.00
## Median :4.445e+09   Median : 10.00
## Mean     :4.848e+09   Mean    : 14.69
## 3rd Qu.:6.962e+09   3rd Qu.: 11.00
## Max.     :8.878e+09   Max.    :157.00
```

```
heart_rate %>%
  select(Time,
          Value)%>%
  summary()
```

For heart_rate

```
##           Time           Value
## Min.      :NA           Min.   : 36.00
## 1st Qu.:NA           1st Qu.: 63.00
## Median :NA           Median : 73.00
## Mean     :NaN          Mean    : 77.33
## 3rd Qu.:NA           3rd Qu.: 88.00
## Max.     :NA           Max.    :203.00
## NA's     :2483658
```

Visualizations and Insights

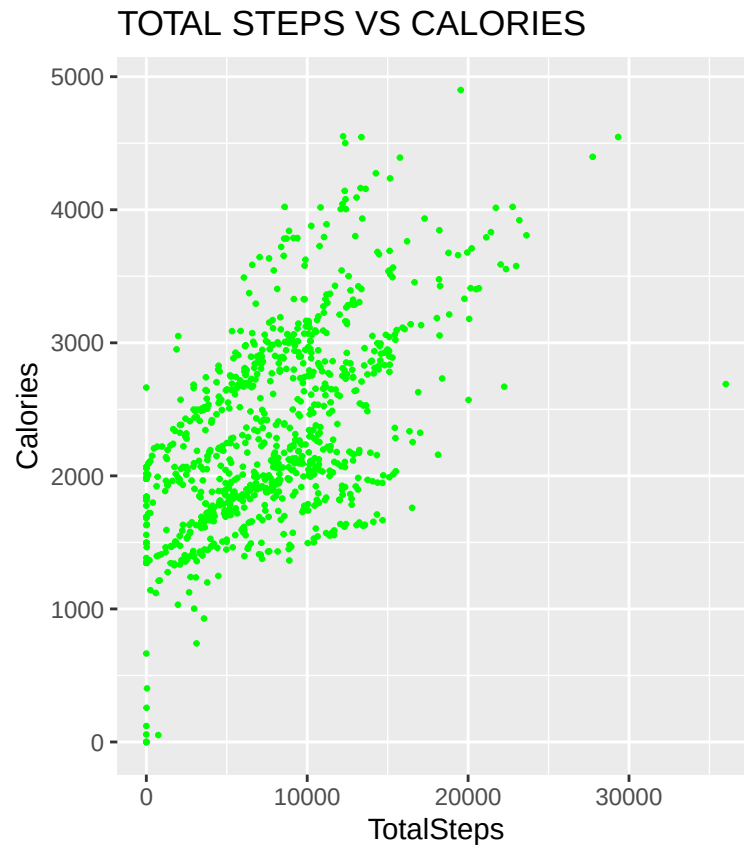
```
install.packages("ggplot2")
```

insatlling ggplot2

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.4'
## (as 'lib' is unspecified)
```

```
library(ggplot2)
```

```
ggplot(data=daily_activity)+ geom_point(mapping = aes(x=TotalSteps, y=Calories),color="green",size=0.5)
  labs(title="TOTAL STEPS VS CALORIES")
```



STEPS Vs CALORIES

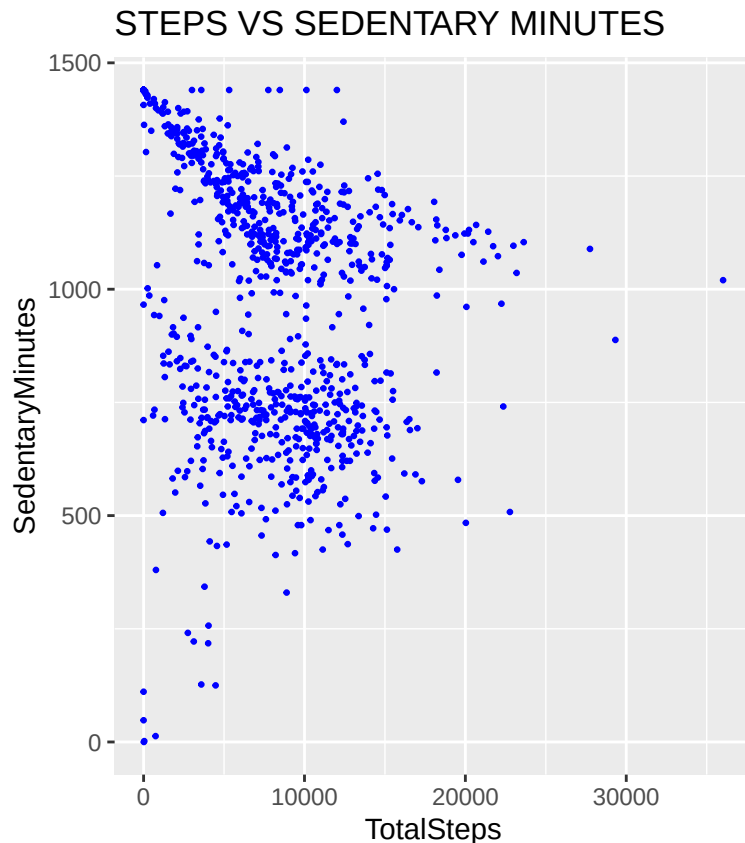
#INSIGHTS DRAWN AND RECOMMENDATIONS **#Steps vs. Calories Relationship:** There's a clear positive correlation between the number of steps taken and calories burned. This reinforces the effectiveness of step tracking in estimating energy expenditure for users.

#Activity Trends: The majority of users fall within the 0–15,000 step range, burning between 1,000–3,000 calories. This suggests that most users engage in moderate activity rather than extreme levels of fitness.

#Identifying High-Performance Users: A few outliers indicate users taking over 20,000 steps and burning more than 4,000 calories. These highly active individuals may be athletes or fitness enthusiasts, presenting opportunities for premium fitness-focused features or targeted marketing.

#STEPS VS SEDENTARY MINUTES

```
ggplot(data=daily_activity)+ geom_point(mapping = aes(x=TotalSteps, y=SedentaryMinutes),color="blue",size=100)+
labs(title="STEPS VS SEDENTARY MINUTES")
```



ANS:There is an inverse relationship between steps taken and sedentary minutes. Users with higher activity levels are less sedentary throughout the day. Most users fall into the low-to-moderate activity range.

How could this help inform the customer segments that we can market to?

ANS:This insight suggests there are many users with high sedentary time and low activity — a prime segment for targeting with motivational fitness content, beginner walking programs, or app reminders for movement breaks.

E.g. position this more as a way to get started in walking more?

ANS:Yes! Bellabeat can position its smart device and app as an ideal starting point for those who are mostly sedentary and want to build healthier habits gradually — using step tracking as a baseline.

Or to measure steps that you're already taking?

ANS:For more active users, Bellabeat can emphasize progress tracking — showing how step increases reduce sedentary behavior. It can reinforce positive feedback and gamify consistency.

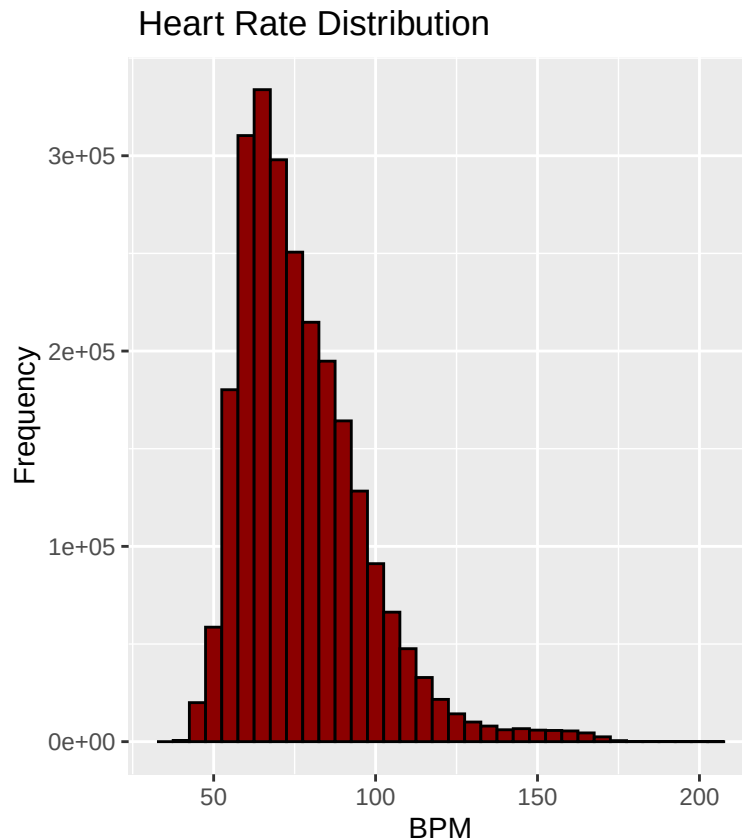
What could these trends tell you about how to help market this product? Or areas where you might want to explore further?

ANS:The clear trend supports two parallel strategies: Encourage inactive users to reduce sedentary time with nudges, walking goals, and micro-challenges.

Market to already active users by offering trend insights, workout summaries, and rewards for consistency.

###HEART RATE DISTRIBUTION

```
ggplot(heart_rate, aes(x = Value))+  
  geom_histogram(binwidth = 5, fill = "darkred", color = "black" )+  
  labs(title = " Heart Rate Distribution", x = "BPM", y = "Frequency")
```



What does this say about the health of customers and what new ideas can be introduced to improve it ? #####ANS:##### What does this say about the health of customers and what new ideas can be introduced to improve it?

ANS: The distribution suggests that most users maintain a **normal resting heart rate** range (60–90 BPM), indicating generally good cardiovascular health during rest. However, the tail of the distribution extending beyond 100 BPM implies that some users experience elevated heart rates, possibly due to stress, poor conditioning, or intense workouts.

Ideas to improve user health:

1. **Heart Rate Alerts:** Introduce heart rate threshold notifications to alert users during periods of elevated BPM.
2. **Stress-Relief Suggestions:** For users with consistently high resting heart rates, Bellabeat could recommend breathing exercises, mindfulness activities, or light walking.

3. **Custom Fitness Zones:** Based on BPM patterns, suggest personalized heart-rate training zones (e.g., fat burn, cardio, peak), improving workout targeting and effectiveness.
4. **Recovery Monitoring:** Help users understand when their heart rate returns to resting levels post-activity — a marker of improved fitness.

```
combined_data <- merge(sleep_day, daily_activity, by="Id")
```

Merging these two datasets together

```
n_distinct(combined_data$Id)
```

Take a look at how many participants are in this data set.

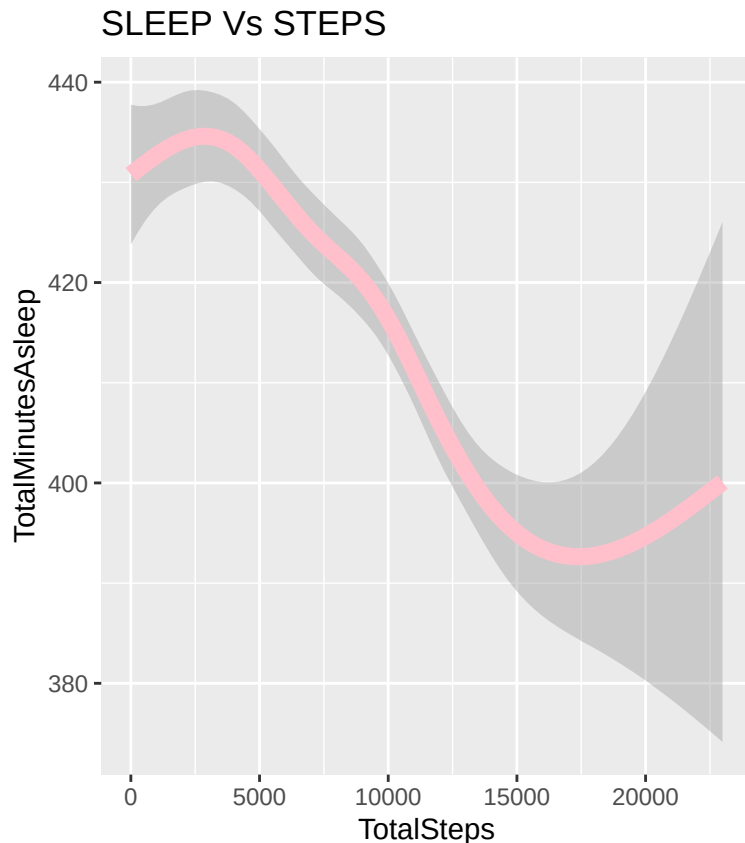
```
## [1] 24
```

Now you can explore some different relationships between activity and sleep as well.

```
ggplot(data = combined_data) +  
  geom_smooth(mapping = aes(x = TotalSteps, y = TotalMinutesAsleep), color = "pink", size = 3)+  
  labs(title = "SLEEP Vs STEPS")
```

For example, do you think participants who sleep more also take more steps or fewer steps per day? Is there a relationship at all? How could these answers help inform

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.  
## i Please use `linewidth` instead.  
## This warning is displayed once every 8 hours.  
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was  
## generated.  
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```



INSIGHTS DRAWN AND RECOMMENDATIONS

1.

There is a **slight inverse relationship** between total steps taken and sleep duration. This might suggest that moderately active individuals may sleep slightly less — possibly due to busy or highly mobile lifestyles.

2.

Users with very high step counts (above 20,000) show a **small rebound in sleep minutes**, possibly due to physical fatigue requiring more rest. This may suggest recovery sleep patterns following intensive activity.

3.

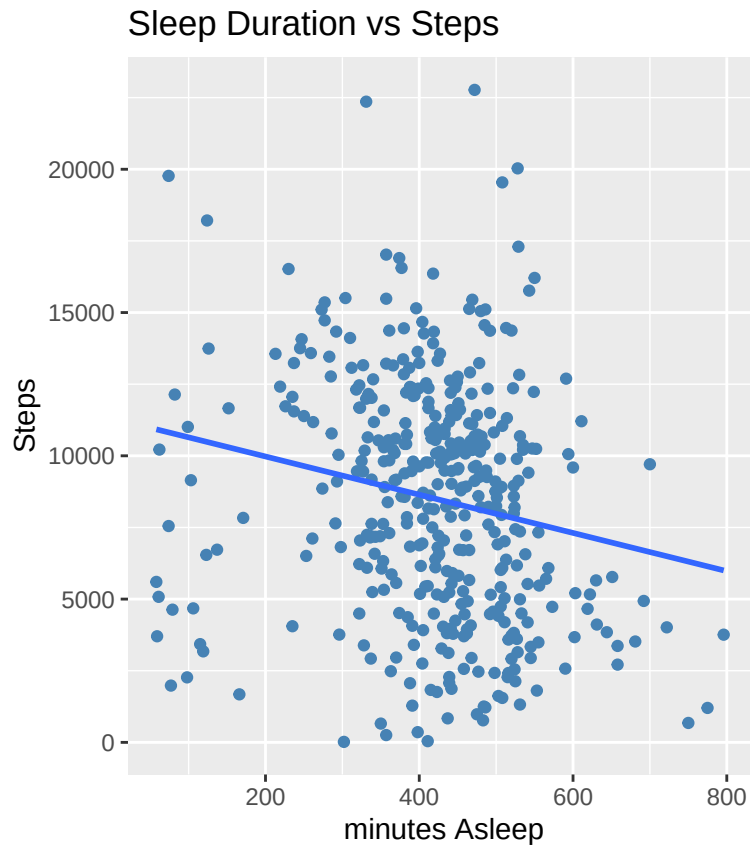
Bellabeat can use this insight to encourage users to: - **Balance activity and rest** by scheduling cooldown evenings or promoting sleep hygiene tips - **Track sleep recovery** following long activity days - Explore “**smart sleep**” coaching, adjusting bedtime recommendations based on user movement data

This cross-analysis supports marketing Bellabeat as a **complete wellness solution** — not just for fitness tracking, but for encouraging healthy rest cycles.

```
activity_sleep <- merge(daily_activity, sleep_day, by.x = c("Id", "ActivityDate"), by.y = c("Id", "SleepDate"))
ggplot(activity_sleep, aes(x= TotalMinutesAsleep, y = TotalSteps))+
  geom_point(color= "steelblue")+
  geom_smooth(method = "lm", se = FALSE)+
  labs(title = "Sleep Duration vs Steps", x = "minutes Asleep", y = "Steps")
```

SLEEP VS ACTIVITY

```
## `geom_smooth()` using formula = 'y ~ x'
```



INSIGHTS DRAWN AND RECOMMENDATIONS

1. The trend line shows a **mild negative correlation** between sleep duration and daily steps. This suggests that users who sleep more than average may be slightly less active — potentially due to lifestyle habits or recovery needs.
2. Most users fall within a **moderate sleep range (5–8 hours)** and show diverse step activity within this bracket, indicating that other factors (motivation, goals, occupation) drive activity levels.
3. Bellabeat can:
 - Introduce **smart sleep-activity balance tracking**, guiding users to avoid oversleeping or inactivity.
 - Push **morning movement prompts** to users with long sleep durations.
 - Position the app as a tool to help users **build consistent patterns**, optimizing both sleep and physical movement.

This analysis supports Bellabeat's role as a lifestyle enhancer — not just a tracker.